SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards *(BL2,BL6)*

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

*I **Antony Magrin (192572103)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
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Annexure A

Industry Problem Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.
2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisychained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for costefficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical threelayer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.
5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department’s network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and wellexecuted implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or nonfunctional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools

Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

1. Star Topology for a 20-Person Startup

A star topology is one of the most suitable network designs for a small startup with approximately 20 employees. In a star topology, every computer, printer, IP phone, wireless access point, and other network device is connected individually to a central networking device, usually a switch. This central switch manages communication between all connected devices.

For a 20-person startup, a 24-port or 48-port managed Ethernet switch can be installed at the center of the network. A 24-port switch is sufficient for connecting 20 employee systems while leaving additional ports for printers, access points, IP phones, servers, and future expansion. A managed switch is preferred because it provides features such as VLAN configuration, port monitoring, traffic management, security controls, and troubleshooting capabilities.

Each employee's computer can be connected to the central switch using Cat6 UTP Ethernet cables. Cat6 cables can support Gigabit Ethernet over typical office cable distances and provide sufficient bandwidth for activities such as web browsing, cloud applications, file sharing, video conferencing, and database access.

The central switch can be connected to a router or firewall that provides internet connectivity. The router can perform functions such as DHCP, NAT, routing, and basic network security. A firewall can inspect incoming and outgoing traffic and prevent unauthorized access to the internal network. A wireless access point can also be connected to the central switch. This allows employees to connect laptops, tablets, and smartphones to the company network without requiring physical cables.

Working of the Network

The operation of the star topology is straightforward. When an employee sends data to another employee, the data first reaches the central switch. The switch checks the destination MAC address and forwards the data to the appropriate device. If the destination is outside the local network, the switch forwards the traffic toward the router or firewall.

For example, if Employee A wants to access a company server, the request travels from Employee A's computer to the central switch and then to the server. If Employee A wants to access a website, the traffic travels from the computer to the switch, then to the firewall/router, and finally to the internet.

Advantages

1. Easy installation: Each device has an individual connection to the central switch.
2. Easy troubleshooting: Network administrators can identify the faulty port or cable quickly.
3. Fault isolation: Failure of one workstation cable affects only that workstation.
4. Good performance: Dedicated Ethernet connections provide reliable communication.
5. Easy expansion: Additional devices can be connected to unused switch ports.
6. Centralized management: The managed switch allows administrators to monitor network traffic.
7. Improved security: VLANs and port security can be implemented.
8. Suitable for modern Ethernet: Star topology works well with Gigabit and higher-speed Ethernet.

Limitation

The main disadvantage is that the central switch represents a single point of failure. If the switch completely fails, all connected devices may lose network connectivity. This can be reduced by keeping a spare switch or using redundant switches for critical environments.

Conclusion

Therefore, a star topology is the best choice for a 20-person startup because it provides a good balance between cost, performance, reliability, manageability, and scalability. It is simple enough for a small organization while still supporting future growth.

2. Mesh Topology for a Banking Network

A banking organization handles highly sensitive financial transactions and requires continuous network availability. Even a short network interruption can affect ATM services, online banking, payment processing, branch communication, and database access. Therefore, a network with high redundancy and fault tolerance is required.

A mesh topology, particularly a partial mesh, is appropriate for the core network of a banking organization. In a full mesh topology, every important network device has a direct connection to every other device. In a partial mesh, only critical devices are interconnected through multiple redundant paths.

A banking network can have multiple core routers, Layer 3 switches, firewalls, and data-center systems. Critical devices can be connected through redundant fiber-optic links. If one link fails, network traffic can use another available path.

Network Components

The major components can include:

- Core routers
- Layer 3 core switches
- Enterprise firewalls
- Fiber-optic links
- Redundant internet connections
- Distribution switches
- Database servers
- Application servers
- Backup servers
- UPS systems
- Redundant power supplies

Use of Fiber Optic Cable

Fiber-optic cables are highly suitable for the banking core network because they provide high bandwidth and low latency. They are also resistant to electromagnetic interference.

Single-mode fiber can be used for long-distance communication between buildings or branches, while multimode fiber can be used for shorter distances within suitable data-center environments.

Redundant Internet Connectivity

Banks can use two or more internet service providers. Multiple enterprise routers can connect to different providers. BGP can be used for redundant external connectivity, while routing protocols such as OSPF can be used within the internal network.

For example, if Internet Service Provider A becomes unavailable, traffic can be redirected through Internet Service Provider B.

Fault Tolerance

Suppose Router A is connected to Router B and Router C. If the link between Router A and Router B fails, traffic can automatically travel through Router C.

Similarly, if one core switch fails, redundant connections can allow traffic to reach the destination through another core switch.

This prevents a single component failure from completely stopping banking operations.

Advantages

1. Very high reliability
2. Multiple alternative communication paths
3. Excellent fault tolerance
4. Automatic traffic rerouting
5. High bandwidth using fiber
6. Suitable for mission-critical applications
7. Reduced impact of individual link failures
8. Supports redundant internet connections

Limitation

The major disadvantage is the high cost and complexity. Multiple routers, switches, firewalls, cables, power supplies, and monitoring systems are required. Network configuration and maintenance also require highly skilled administrators.

Conclusion

A partial mesh topology is highly suitable for banking networks because it provides multiple paths, redundancy, fault tolerance, and continuous connectivity. A full mesh can be used for a very small number of critical devices, while a partial mesh is more practical for large banking infrastructure.

3. Bus Topology for a Small Classroom Lab

A small classroom laboratory with limited financial resources may require an inexpensive networking solution. Traditionally, a bus topology was used because all computers shared a single communication backbone.

In a traditional bus network, all computers are connected to one main cable. The cable acts as a shared communication medium. Data transmitted by one computer travels along the backbone, and the appropriate destination computer receives the data.

A traditional Ethernet bus network commonly used coaxial cable, such as 50-ohm RG-58 cable, along with BNC connectors, T-connectors, and terminators.

Components

A traditional bus network requires:

- Coaxial backbone cable

- BNC connectors
- BNC T-connectors
- 50-ohm terminators
- Network interface cards
- Computers

The terminators are installed at both ends of the backbone cable. They prevent signal reflections that could interfere with communication.

Working

When a computer sends data, the signal travels along the common backbone cable. All connected devices can detect the signal, but only the device whose address matches the destination accepts the data.

Since multiple computers share the same communication medium, simultaneous transmission can cause collisions.

Traditional Ethernet bus networks used CSMA/CD mechanisms to deal with collisions.

Advantages

1. Low initial hardware cost
2. Less cabling compared with individual connections
3. No central switch required in the traditional implementation
4. Simple concept
5. Suitable for educational demonstrations

Disadvantages

1. A backbone cable failure can affect the entire network.
2. Troubleshooting can be difficult.
3. Network performance decreases with increased traffic.
4. Collisions can occur.
5. Adding devices can become difficult.
6. The topology has limited scalability.
7. Traditional bus Ethernet is obsolete in modern enterprise networks.

Modern Alternative

Although bus topology is useful for explaining networking concepts, it is generally not recommended for modern classroom networks. An inexpensive unmanaged Ethernet switch with Cat5e or Cat6 cables provides better reliability and performance.

The modern implementation would therefore use a small star topology while maintaining a low budget.

Conclusion

Bus topology can be considered for a low-cost educational environment or for demonstrating traditional networking concepts. However, for an actual classroom network, a small switched star topology is a better practical solution because it provides higher reliability, better performance, easier troubleshooting, and easier expansion.

4. Hybrid Topology for a Large University Campus

A university campus may contain thousands of users distributed across several buildings. These buildings may include classrooms, laboratories, libraries, administrative offices, hostels, research centers, and lecture halls.

Because of the large size and different requirements of each area, using only one topology would not be efficient. Therefore, a hybrid topology is the most appropriate solution.

A hybrid topology combines multiple topology types. For example, the university can use a partial mesh or ring at the campus backbone, a star topology inside buildings, and a hierarchical structure between the core, distribution, and access layers.

Campus Core

The campus core can consist of high-performance Layer 3 switches and routers connected using fiber-optic cables.

Redundant connections can be created between core devices. This ensures that if one fiber link fails, another path can carry the traffic.

For example:

Building A → Core Switch 1

Building A → Core Switch 2

If the connection to Core Switch 1 fails, Building A can continue communicating through Core Switch 2.

Distribution Layer

Each building can have one or more distribution switches. These switches aggregate traffic from different floors and connect the building to the campus backbone.

The distribution layer provides:

- Routing
- VLAN management
- Policy enforcement
- Traffic control
- Redundancy

Access Layer

The access layer connects end-user devices such as:

- Desktop computers
- Laptops
- Printers
- IP phones
- CCTV cameras
- IoT devices
- Wireless access points

Cat6 Ethernet cables can be used for wired devices.

Wireless Connectivity

Wireless access points can be installed throughout classrooms, libraries, hostels, and common areas. PoE-enabled switches can supply both data connectivity and electrical power to compatible access points.

Scalability

One of the biggest advantages of a hybrid topology is scalability. If the university constructs a new building, it can install a new distribution switch and connect it to the campus backbone using fiber.

The existing buildings do not need to be completely redesigned.

Advantages

1. Highly scalable
2. Supports thousands of users
3. Provides backbone redundancy
4. Easy building-level management
5. Supports wired and wireless devices
6. Provides better fault isolation
7. Allows different technologies in different areas
8. Supports future expansion
9. Provides centralized network management
10. Suitable for large geographical areas

Limitation

The main disadvantages are higher installation cost and greater design complexity. Fiber infrastructure, redundant switches, routers, wireless access points, and network management systems require significant investment.

Conclusion

A hybrid topology is the most suitable choice for a university campus because it combines the advantages of several topology types. It provides high availability at the backbone, easy management within buildings, and scalability for future campus expansion.

5. VLAN-Based Hierarchical Network for a Multi-Department Company

A company containing departments such as Human Resources, Finance, Sales, Marketing, and IT requires a structured network. If all employees are placed on the same network, unnecessary broadcast traffic may increase and sensitive departmental information may be more difficult to protect.

A hierarchical network architecture combined with VLANs can solve these problems.

The network can be divided into three major layers:

1. Access Layer
2. Distribution Layer
3. Core Layer

Access Layer

The access layer consists of Layer 2 switches. Employee computers, printers, IP phones, and other devices are connected to these switches.

Separate VLANs can be created for each department.

For example:

Department	VLAN
HR	VLAN 10
Finance	VLAN 20
Sales	VLAN 30
IT	VLAN 40
Management	VLAN 50

Devices belonging to the same VLAN can communicate within that logical network.

Distribution Layer

The distribution layer aggregates traffic from access switches. It can contain Layer 3 switches that perform inter-VLAN routing.

For example, an employee in the Sales department may need to access an application server located in the IT network. The Layer 3 switch can route the traffic between the appropriate VLANs.

Access-control policies can determine which departments are allowed to communicate with each other.

Core Layer

The core layer provides high-speed connectivity between major network segments. Highperformance Layer 3 switches can be used to forward traffic efficiently.

The core layer can also connect the internal company network to data-center resources and other major network services.

Internet Connectivity

A router/firewall connects the internal network to the internet.

The firewall can provide:

- Traffic filtering
- NAT
- Access control
- Threat prevention
- VPN connectivity
- Internet security

Security Benefits of VLANs

VLANs provide logical separation between departments.

For example, Finance systems can be placed in a separate VLAN from Sales systems. Accesscontrol rules can then restrict unauthorized communication between them.

This is particularly useful because financial information and employee records may require stronger access restrictions.

IP Addressing

Each VLAN can have its own IP subnet.

For example:

- HR → 192.168.10.0/24
- Finance → 192.168.20.0/24
- Sales → 192.168.30.0/24
- IT → 192.168.40.0/24

This makes network administration and troubleshooting easier.

Redundancy

For important departments, redundant links can be implemented between switches. Multiple uplinks can provide backup connectivity.

If one network link fails, traffic can use another available link.

Advantages

1. Improved security
2. Reduced broadcast traffic
3. Better network performance
4. Easy department-wise management
5. Efficient IP address management
6. Improved scalability
7. Supports inter-VLAN communication
8. Easy troubleshooting

9. Supports access-control policies
10. Allows future departmental expansion

Conclusion

A hierarchical network with VLAN segmentation is an effective solution for a multi-department company. It provides centralized management, improved security, efficient communication, reduced broadcast traffic, and scalability. By combining Layer 2 switches, Layer 3 switches, routers, firewalls, VLANs, and proper IP addressing, the company can build a reliable and secure enterprise network.

Overall Comparison

Problem	Recommended Topology	Main Technology	Major Benefit
20-person Startup	Star	Managed Switch + Cat6	Easy management
Banking Network	Partial Mesh	Fiber + Redundant Routers	High availability
Small Classroom	Bus / Modern Star	Coaxial or Ethernet	Low cost
University Campus	Hybrid	Fiber + Core/Distribution/Access	Scalability
Multi-Department Company	Hierarchical + VLAN	L2/L3 Switches + VLANs	security